

Akshaj Kashyap

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EDUCATION

University of California, Santa Barbara

Santa Barbara, CA

B.S. Computer Science | Second-year undergraduate | GPA: 3.76

Expected June 2028

- Admitted to UCSB's Early Research Scholars Program (ERSP) for 2026–27.
- **Coursework:** Data Structures and Algorithms, Computer Architecture, Probability & Statistics, Systems Programming

EXPERIENCE

PLAXCO Lab

Aug. 2025 – Present

Undergraduate Researcher

Santa Barbara, CA

- Processed 44 electrochemical exports (11 datasets $\times$ two electrodes $\times$ two methods) into concentration metrics and plots while continuing to refactor SACMES into reusable Python analysis code.

Techions

May 2024 – Aug. 2024

Technology Intern

San Jose, CA

- Developed a computer vision pipeline using Roboflow, depth estimation, and Segment Anything to identify vegetation near cellular infrastructure for wildfire-risk screening.

Handshake AI Fellowship

Jan. 2026 – Present

AI Evaluation Fellow

Remote

- Completed 20+ rubric-based evaluations of AI-generated image edits, scoring realism and coherence and documenting judgments with structured written justifications.

PromptShop

Jan. 2026 – Present

Software / Prompt Engineering Contributor

Remote

- Designed and iterated five prompt-driven content workflows, evaluating outputs on 30+ structured test cases.

PROJECTS

CUDA Transformer Runtime | C++20, CUDA, cuBLAS, CMake | [GitHub](#)

- Developed a C++20/CUDA transformer runtime spanning BPE tokenization, LLaMA-style decoder layers, KV-cached greedy generation, and custom kernels checked against independent CPU references.
- Passed 21/21 CTest checks with 2.563×10^{-5} maximum CPU/CUDA logit error and generated a 64-token continuation at about 266.8 tokens/s on an RTX 3050 Laptop GPU.

Causal Uplift Experimentation | Python, scikit-learn, FastAPI | [GitHub](#)

- Compared S-, T-, and random-forest uplift learners using cross-fitting, treatment-aware ranking, bootstrap uncertainty, sensitivity analysis, and policy simulation.
- Packaged a frozen policy artifact and staging API; a synthetic prospective-trial simulation measured +4.36 percentage points of conversion lift, while promotion remained on hold pending real randomized validation.

Molecular Property Prediction with GNNs | PyTorch Geometric, RDKit, FastAPI | [GitHub](#)

- Compared GCNs, GINs, and Morgan-fingerprint baselines on an ESOL scaffold split; across three seeds, the GCN reached 1.3395 $\pm$ 0.0738 test RMSE versus 1.8480 $\pm$ 0.0214 for random forest.
- Implemented validation-only model promotion and FastAPI/Docker serving with 142 tests; withheld confidence estimates after ensemble disagreement failed to rank errors, reporting applicability context instead.

LEADERSHIP

SciTrek

Aug. 2025 – Present

Mentor

Santa Barbara, CA

- Helped develop a bioinformatics module used by 100+ K–12 students, translating technical concepts into accessible STEM lessons.

C.S.E.P. (F.U.S.E.)

Sep. 2025 – Present

Mentor

Santa Barbara, CA

- Selected as the program's only undergraduate mentor; helped develop and deliver a STEM outreach module across 9+ classrooms and 250+ students.

TECHNICAL SKILLS

Languages: Python, C++, SQL

ML/Libraries: PyTorch, PyTorch Geometric, scikit-learn, RDKit, NumPy, pandas

ML Systems: CUDA, cuBLAS, CMake, FastAPI

Tools/Platforms: Docker, GitHub Actions, Git, Linux